

# Juriel Garcia S

He/Him • Toronto (hybrid-ready)

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## Profile

Front-end engineer focused on interactive 3D experiences in the browser. I design and implement performant 3D viewers and immersive UIs using TypeScript and WebGL (three.js), with a strong background in game development and real-time rendering. I enjoy translating complex visuals into robust, reusable components and collaborating with UX and backend teams to deliver production-quality, cross-browser applications.

## Key Skills

- **Languages / Frameworks:** TypeScript, JavaScript (ES6+), React, Next.js, HTML5, CSS3
- **3D / Graphics:** three.js, WebGL, GLSL shaders, GLTF/FBX/OBJ pipelines, PBR lighting
- **Performance:** GPU-driven rendering, LOD, lazy-loading, binary formats, texture streaming, frustum & occlusion culling
- **Tools / Engines:** Blender, Unity (developer pipelines), FBX & GLTF tooling
- **Backend / Cloud:** REST/GraphQL APIs, S3/Cloud Storage for assets, authentication integrations
- **Developer Practices:** modular/typed code (TS), testing, code reviews, Git
- **Soft skills:** close collaboration with UX/design, strong documentation, mentoring and cross-discipline communication

## Selected Projects

- **AT4D - Personal** [↗](#) 2019 – 2022
  - Built custom 3D scenes using **TypeScript**, **three.js** and **React** that loads GLTF/GLB assets, supports animations, and dynamic behaviours.
  - Designed custom PBR shaders and post-processing passes to improve perceived fidelity while keeping GPU & CPU budgets low for web delivery.
  - Implemented progressive asset streaming (binary GLB + compressed textures), LOD, and frustum culling to maintain 60fps for heavy scenes.
  - Worked with UX to craft camera controls, spatial navigation, and touch/gesture interactions suitable for desktop and tablet use.
- **Willow Games Site** [↗](#) 2020 – 2024
  - Built custom 3D scenes using **TypeScript**, **three.js** and **React** that loads GLTF/GLB assets, supports animations, and dynamic behaviours.

## Experience

- **Contract Game Developer** 2024 - 2025
  - Implemented visualization tools for DOTS-based systems, including dependency graph viewers and dynamic component editors.
  - Built real-time networked prototypes and integrated rendering pipelines focused on stable, high-performance playback.
- **Partner & Game Developer - Willow Games** 2020 – 2024
  - Led creation of a React + AWS web platform powering marketing sites and asset backends; worked closely with artists to serve 3D previews and demos.
  - Designed tooling for artists and engineers to streamline asset delivery to the web app.
- **Partial Load Professor - Humber College** 2022 – 2024
  - Taught Game Programming and Object-Oriented Design; introduced students to 3D math, rendering basics, and game-engine workflows.

## Education

- **Humber College, Toronto** 2017 – 2021
  - Game Programming - Advanced Diploma